TK-TD modelling I

Bioaccumulation Modelling

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June 2026

Bioaccumulation example

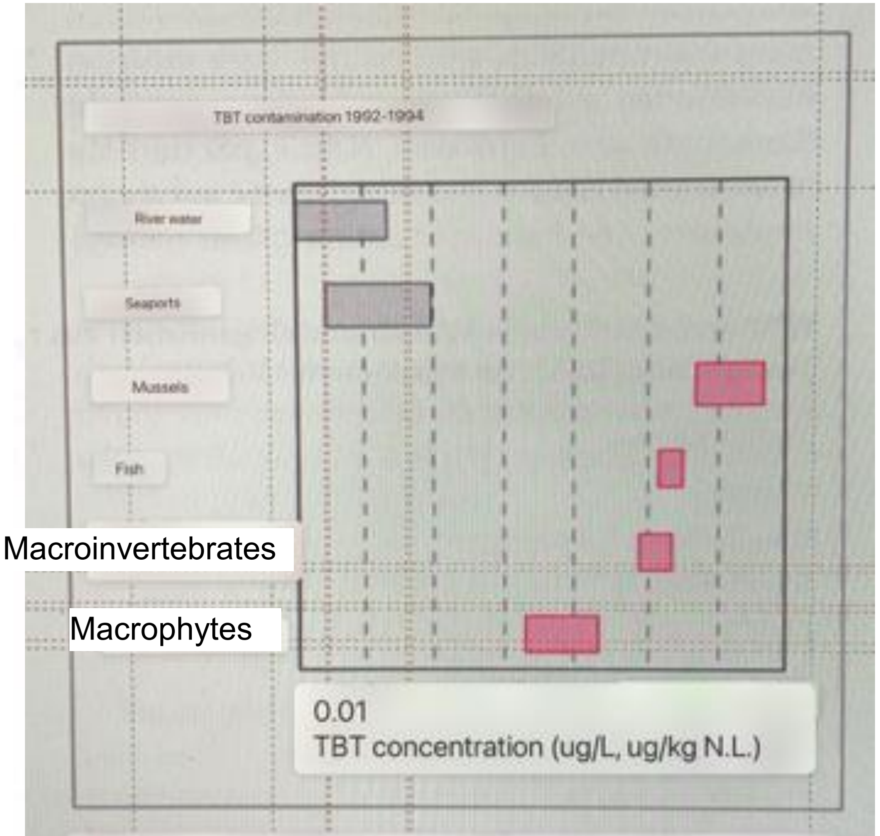


Fig. 7.1 Distribution of tributyl tin chloride (TBT) in the period 1992-1994 in the harbour water of Lake Lucerne (Switzerland) and in various organisms occurring in the harbour.

Figure from: Fent „Ökotoxikologie" Thieme Verlag

// Distribution of TBT concentrations in harbour water and different organism groups

// TBT concentration in organisms higher than in the environment

// TBT concentration varies among organism groups

Exposure and uptake of chemicals

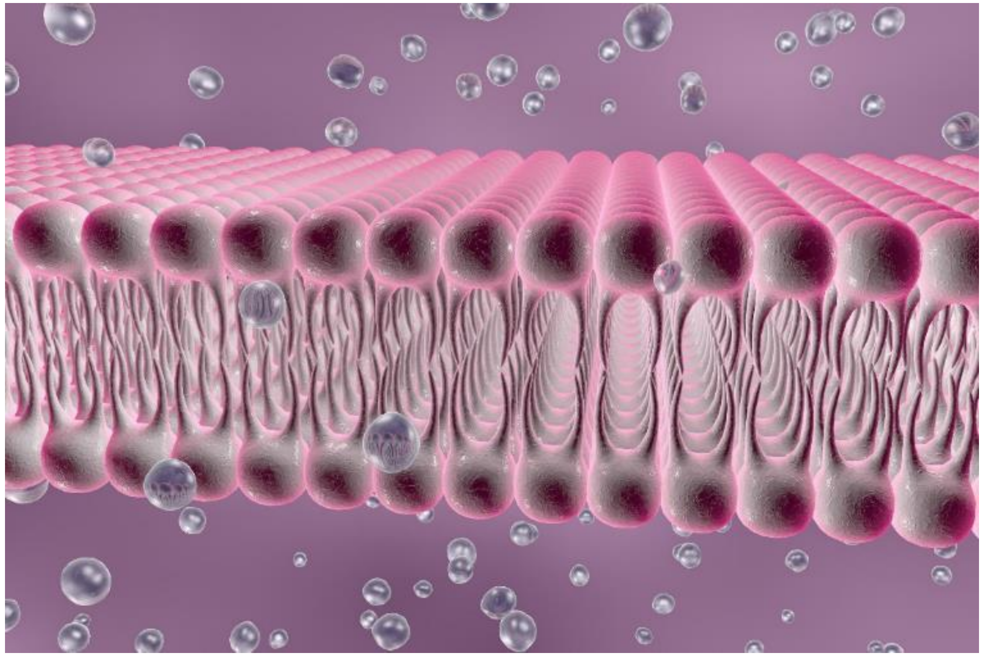
// Uptake into the organism results in internal concentrations

// Depends essentially on three factors:

1. Exposure in the organism's environment
2. Properties of the chemical
3. Properties of the organism

// Internal concentration is the sum of uptake and loss processes

Uptake and elimination processes



// Uptake

- Passive/diffusion via membrane (e.g. skin)
- Active respiration (air, water)
- Feeding processes

// Loss/excretion

- Back-diffusion
- Respiratory loss, excretion
- Metabolization, degradation

Definitions

Bioaccumulation

Refers to the general process leading to increased xenobiotic concentrations in organisms (internal body concentration) compared to the exposure concentration in the environment.

Bioconcentration

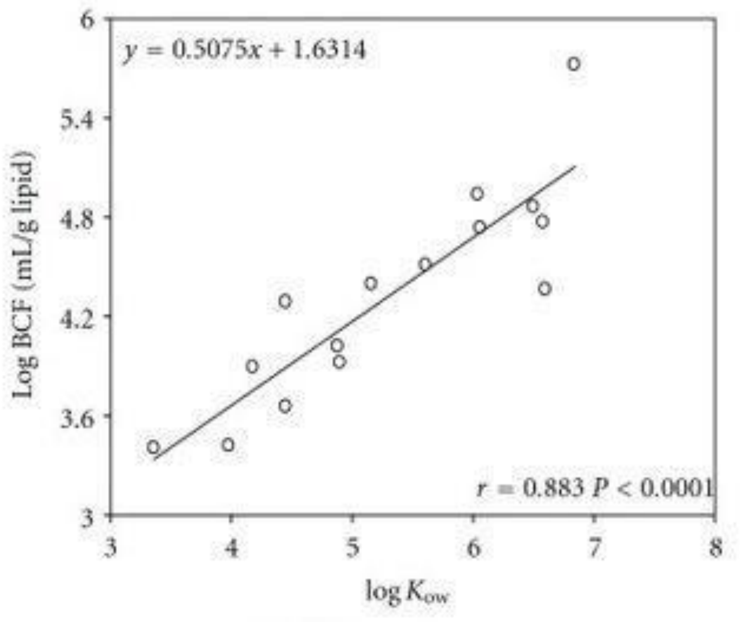
- Uptake of a xenobiotic compound into the organism via **water**
- Uptake pathway via respiratory surfaces or the integument
- Increased body concentrations compared to environmental media (e.g. water)
- Bioconcentration factor (BCF) serves as a measure for accumulation potential

Biomagnification

- Uptake of a xenobiotic compound into the organism via **diet**
- Uptake pathway via gut
- Increased body concentrations compared to concentration in food

Properties of the chemical

Uptake of polycyclic aromatic hydrocarbons in Tilapia



Li & Ran. 2012. DOI: 10.1100/2012/632910

// Influence environmental behavior and exposure

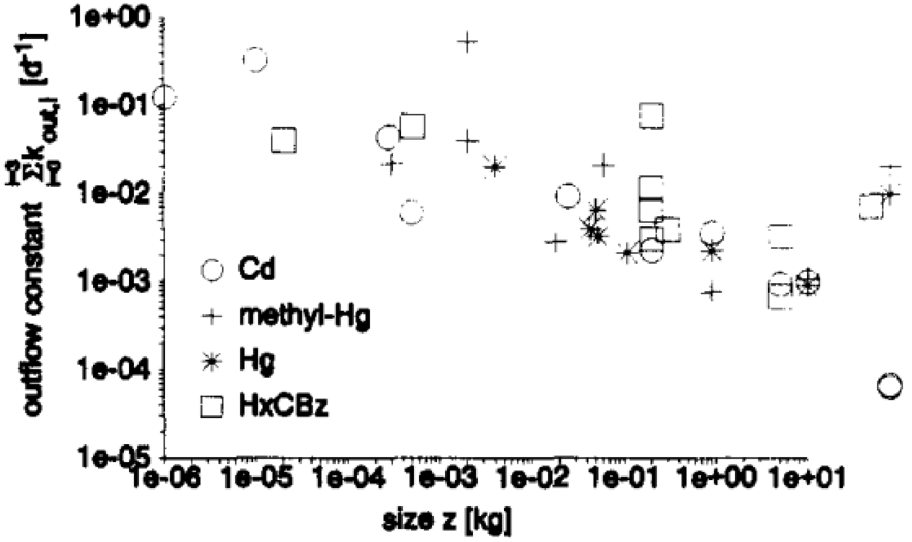
- Emission pathways
- (Abiotic) degradation rate constants
- Partitioning constants - distribution across environmental compartments (e.g. octanol-water; K_{ow})

// Influence uptake into and fate in the organism

- Partitioning constants (body)
- Degradation rate constants in the body

Properties of the organism

Elimination of cadmium, mercury and hexachlorobenzene



Hendriks. 1995. https://doi.org/10.1016/0045-6535(94)00389-C

// Exposure and bioavailability

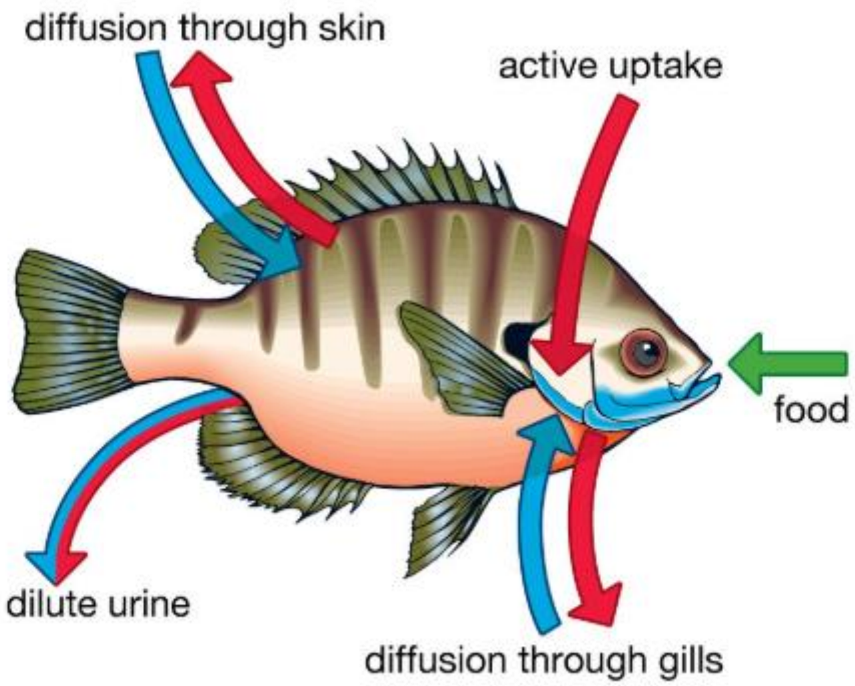
- Habitat, environmental copartments
- Interactions with environment (e.g. respiration and food intake)
- Permeability of skin, gut surface etc.

// Influence uptake into and fate in the organism

- Organism state, internal structuring
- E.g. size, lipid content

Toxicokinetics (TK)

Mechanistically modelling the uptake of chemicals

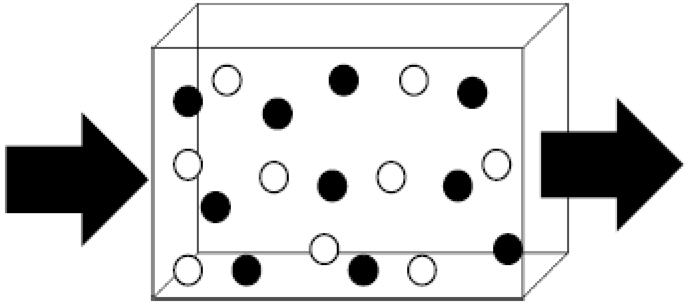


https://duanvanphu.com/how-do-fish-absorb-oxygen/

// Explicitly simulating uptake and elimination pathways

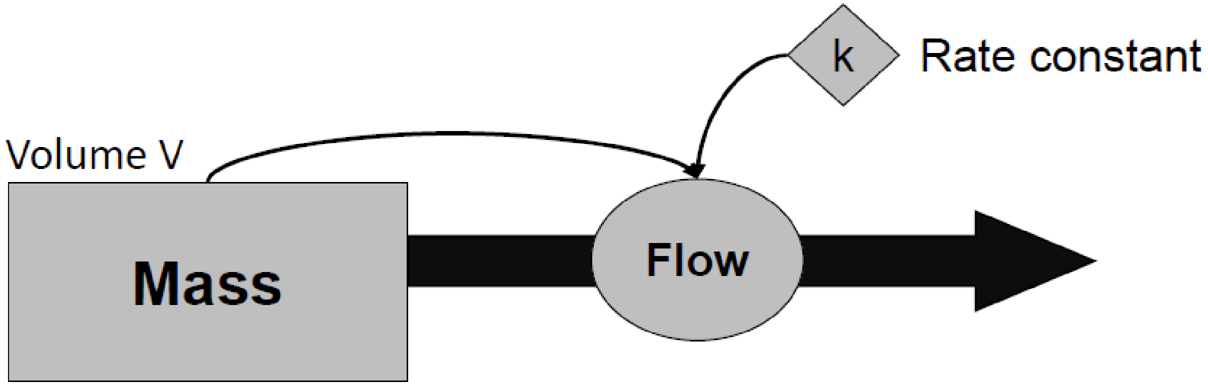
- Diffusion
- Respiration
- Feeding
- Excretion

// In the simplest case an organism is considered as a (homogeneously mixed) compartment



Some modelling essentials

Compartment model



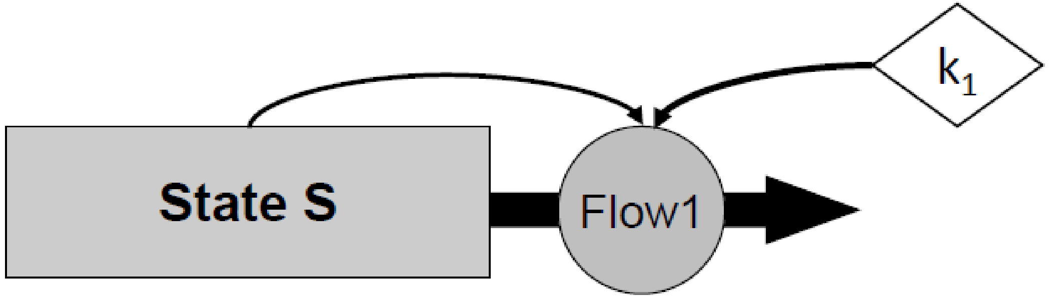
// A compartment is an abstract unit with an associated state

// Environmental compartments are defined as phases or phase mixtures (e.g. water, soil, air, fish)

// Compartments have defined volumes and masses

// The compartment state is changes by flow which determined by a rate constant

Changes in state



// Change in state S (ΔS)

// …in discrete time steps (Δt),

// …by the Flow1

// …determined by the rate constant k_1

$$\Delta S = - Flow_1 \cdot \Delta t$$

From the conceptual model to differential equation

$$\Delta S = - Flow_1 \cdot \Delta t$$

// Change in state S (ΔS) depends on the flow for a discrete a time step (Δt)

$$\frac{\Delta S}{\Delta t} = - Flow_1 = - k_1 \, S$$

// Change in state S (ΔS) per time step (Δt) depends on rate k_1 and S

$$\frac{dS}{dt} = - k_1 \, S\,(t)$$

// For smaller time steps the equation converges towards the differential equation

// This is an ordinary differential equation.

// The change of state S in time depends on the state of S itself

Kinetics – Order of processes

The four most common types of kinetics in ecotoxicological modelling

Kinetics	Explanation	Example ODE
Zero-order kinetics	The change in the state variable is independent of the state variable itself	$\frac{d}{dt}A = k$
First-order kinetics	The change in the state variable linearly depends on the value of one state variable	$\frac{d}{dt}A = kA$
Second-order kinetics (law of mass action)	The change in the state variable linearly depends on the product of two state variables	$\frac{d}{dt}A = kA^2$
Hyperbolic kinetics (Michaelis-Menten kinetics)	The change in the state variable is a hyperbolic function of the value of one state variable	$\frac{d}{dt}A = \frac{kA}{K+A}$

Table by Tjalling Jager

Components of ecological models

From a mathematical view, ecological models consist of five components:

1. Control functions or input variables

- Influence system state externally
- E.g. in an ecotoxicological model: “Which effect does a chemical have?” or in a climate change setting: “How does system state change with increasing temperature?”

2. State variables

- Variables that describes (quantifies) the state of a system
- Choice of state variables is an important step during model development
- E.g. Light: “on/off” or body size : “5cm”
- E.g. in a bioaccumulation model: internal (body) concentration of a chemical

Components of ecological models

3. Mathematical equations

- Represent biological, chemical or physical processes
- Describe relations between control functions and state variables
- Might be formulated as sub-models

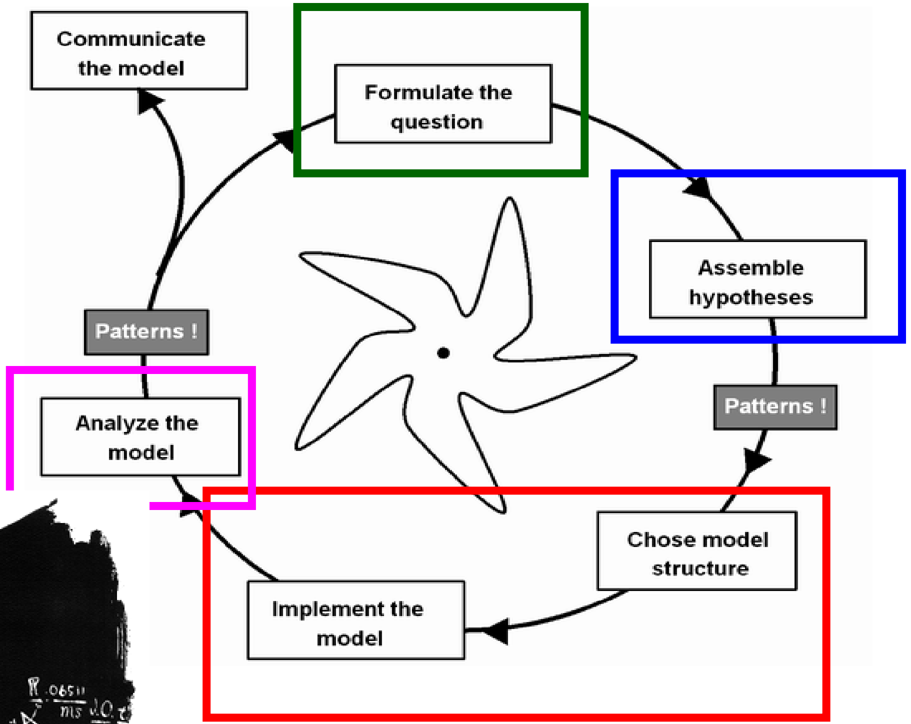
4. Parameter

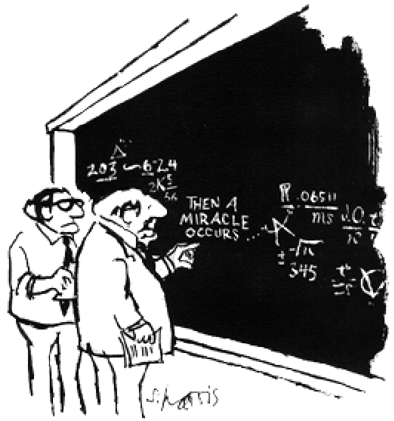
- Coefficients in mathematical equations
- E.g. elimination rate of cadmium in a fish bioaccumulation model

5. Universal constants

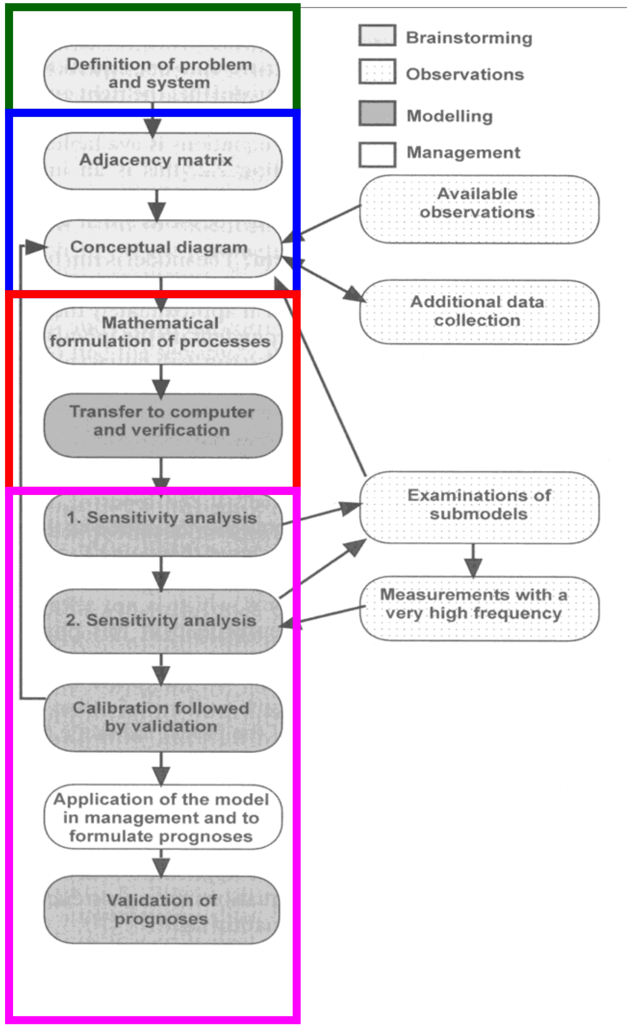
- E.g. gas constant or atomic mass

Conceptual model development





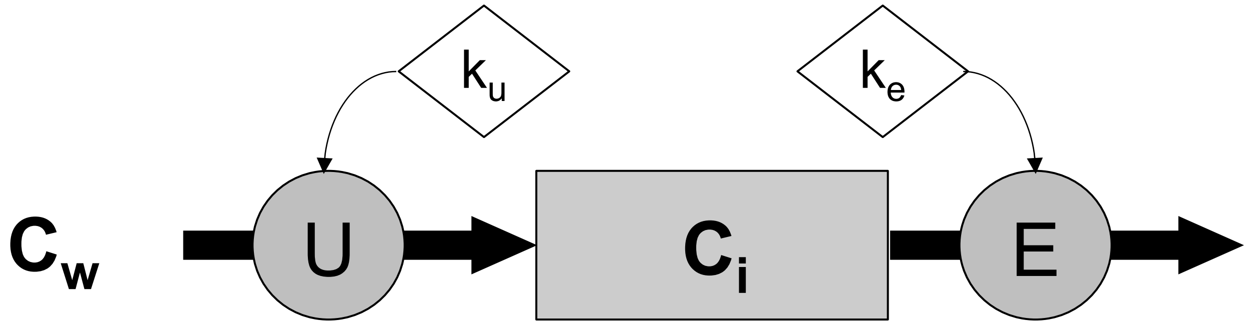
"I think you should be more explicit here in step two."



Back to toxicokinetics

One compartment toxicokinetic (bioaccumulation) model

Simulate uptake and elimination of a chemcial from water into the body



C_w= Exernal (e.g. water) concentration

C_i = Internal concentration

U = Uptake

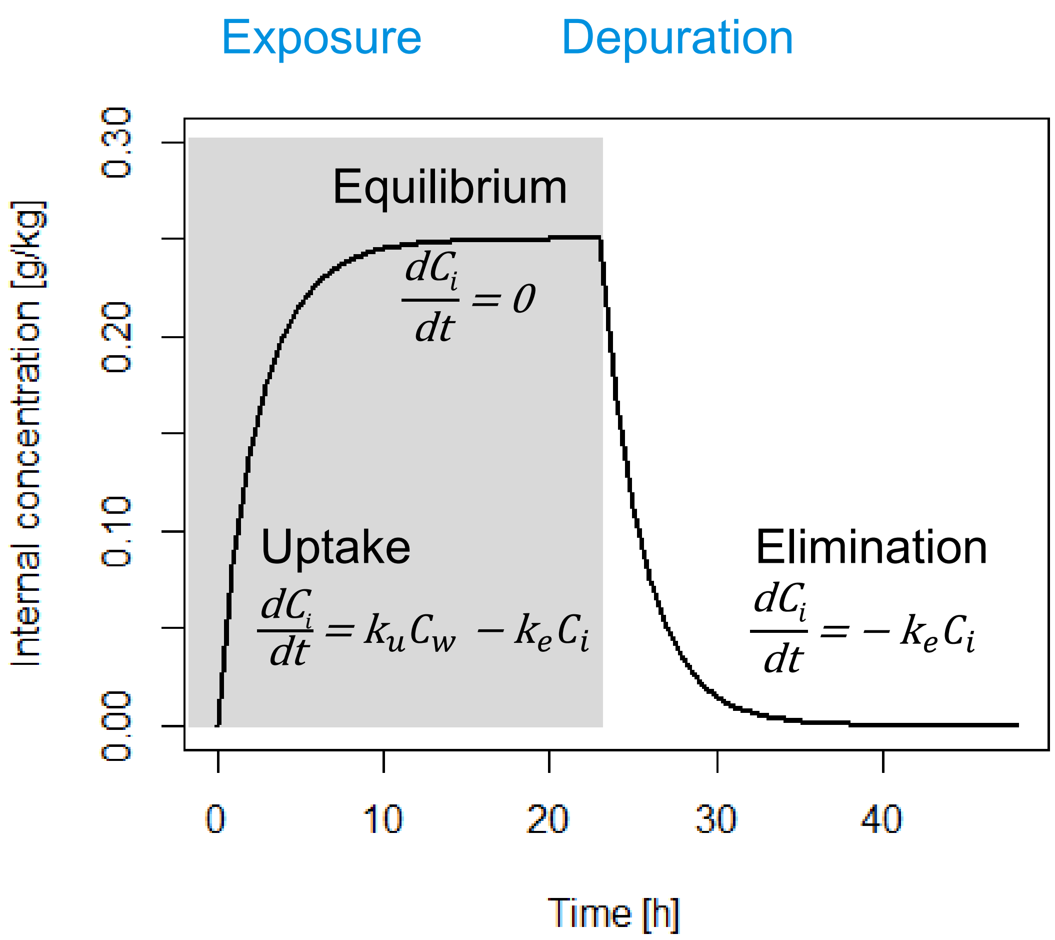
E = Elimination

k_u = Uptake rate

k_e = Elimination rate

$$\frac{dC_i}{dt} = k_u C_w - k_e C_i$$

One compartment model



- // Experiment for quantification of uptake and elimination (depuration) using intermitted exposure
- // During the uptake phase organisms are exposed for a limited period at sublethal concentrations
- // …followed by a non-exposure period where organisms are transferred to clean media

Exercise 1

- // Implement the one-compartment bioaccumulation model (equation on slide 18) in R and present model predictions in a graph (line plot)
- // Try different parameter values for k_u and k_e – does the model behave as expected?
- // Try different exposure concentrations (including an uptake phase and a depuration phase) – does the model behave as expected?

Some useful code examples in R

```
#set your working directory
Setwd(" C:/…" )

#assign a value
x<-1
#assign a vector
x<-c(1:5,7)
# assign a matrix
x <- matrix(data = NA, nrow = 2, ncol = 4)

#loop
for (x in 1:max) {
 # add code here }


#if-else statement
if (x < max) {
 # add code here }
else {
 # add code here }
```

Some useful code examples in R

```
#load data from csv file
data <- read.csv(„datafile.csv", header=TRUE, sep=",")
data <- read.csv(„datafile.csv", header=FALSE, sep=",", skip=3)
```

```
#select a subset of data that fulfills a certain condition e.g. >, < or ==
m<-data[which(data[,2]<0.1),]
```

```
#plot example
plot(data[,1], data[,2], type = "n", xlab = "Concentration [mg/L]",
ylab = "Effect [%]", main = „DATA", cex.axis=1.5, cex.lab=1.5,
cex.main=1.5, ylim=c(0, 100), log="x")
```

Some useful R code

```
#implement and call a function in R (‚desolve ‘ package)
my_function<- function(t, y,  pars, input) {
  with(as.list(c(y, pars)), {
    C <- input(t)
   #change in state
    dstate <- p*C + q*state
  return(list(c(dstate))   })}

#input
t<-c(0:5) # time vector
X <- rep(1,length(t)) #some input values
C <- approxfun(t, X,  rule = 2, method = ″constant″) #interpolation
#initial state variables
yini<- c(state=0...)
#parameters
pars<-c(p=1, q=2...)

#call function, collect outputs
output<- lsoda(y = yini, times = t, func = my_function, pars, input=C)
```

Some useful code examples in R

```
#fit a dose response model, uses 'drc ' package
model.logistic <- drm(m[,2]~m[,1], fct=LL.2(), type="binomial")
 plot(model.logistic, main='logistic model',log='x',xlim=c(0.1,10000),ylim=c(0,1))
   results<- summary(model.logistic)


#install package, load library
install.packages('drc')
library('drc')

#save data in csv file
write.table(results ,sep=',',dec='.', „results.csv “, row.names=FALSE)
```

Exercise 2

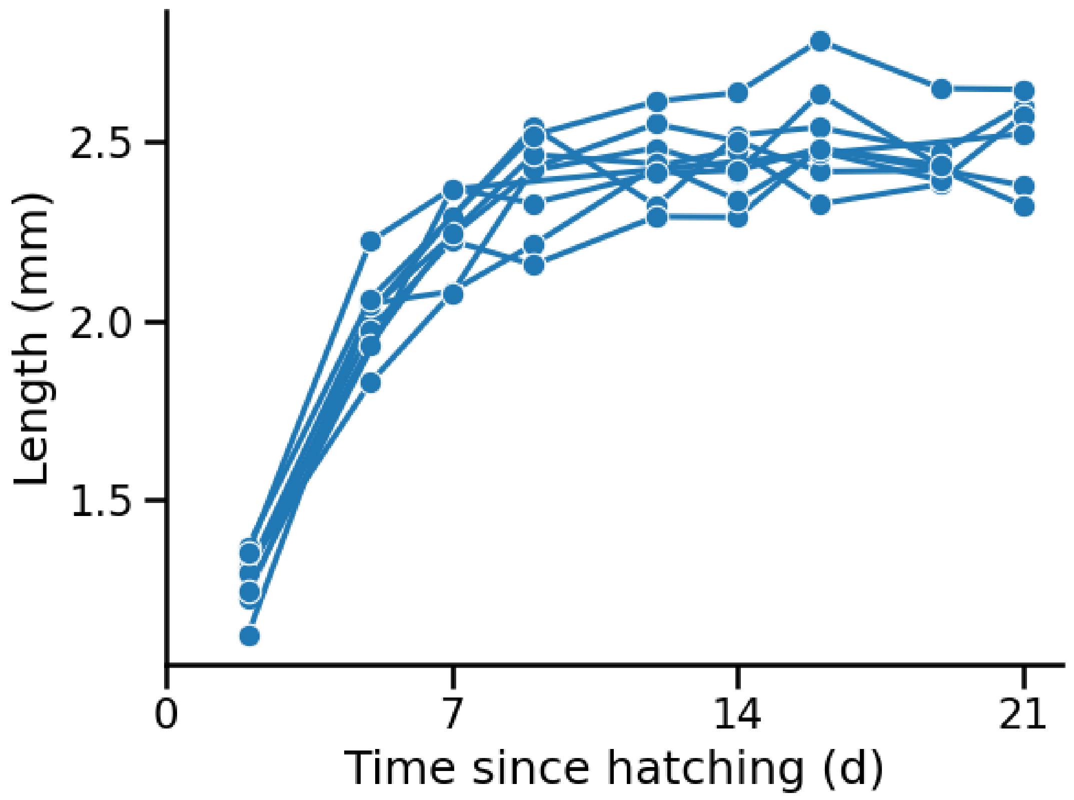
Bioaccumulation of cadmium has experimentally been measured in *Daphnia magna*. Elimination [%] has been determined after transfering daphnids to clean media (for data see the xlsx file).

// Use script from Exercise 1. Curate the data file in an R readable format and read the data from file.
Present both data (dots) and model simulation (line) in one plot

// Estimate the parameters k_u and k_e by trial and error (intially assume $k_u = k_e$)

Some words on parameter estimation

Model fitting example



Figures and slide concept: Simon Hansul

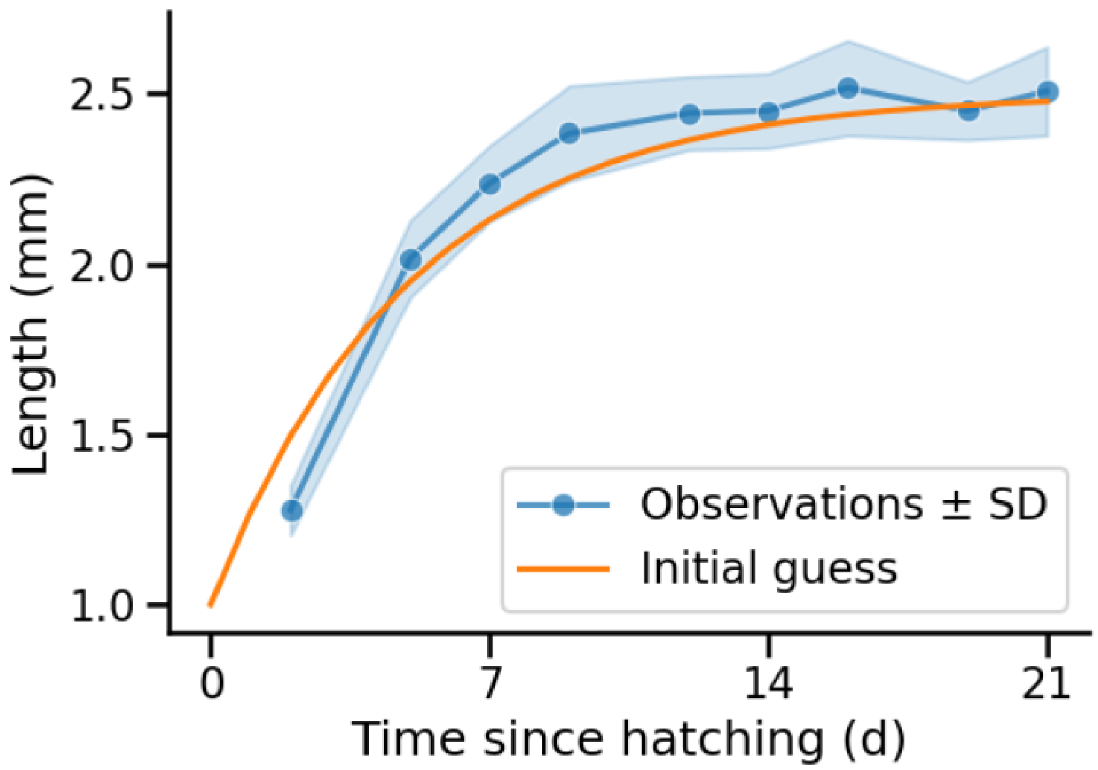
// Toy example: Length of *Daphnia pulex* measured over time

// Model used: von Bertalanffy growth

$$\frac{dL}{dt} = r_B \left(L_\infty - L \right)$$

// with r_B = growth rate (1/d)
L_∞= ultimate length (mm)

Model fitting example



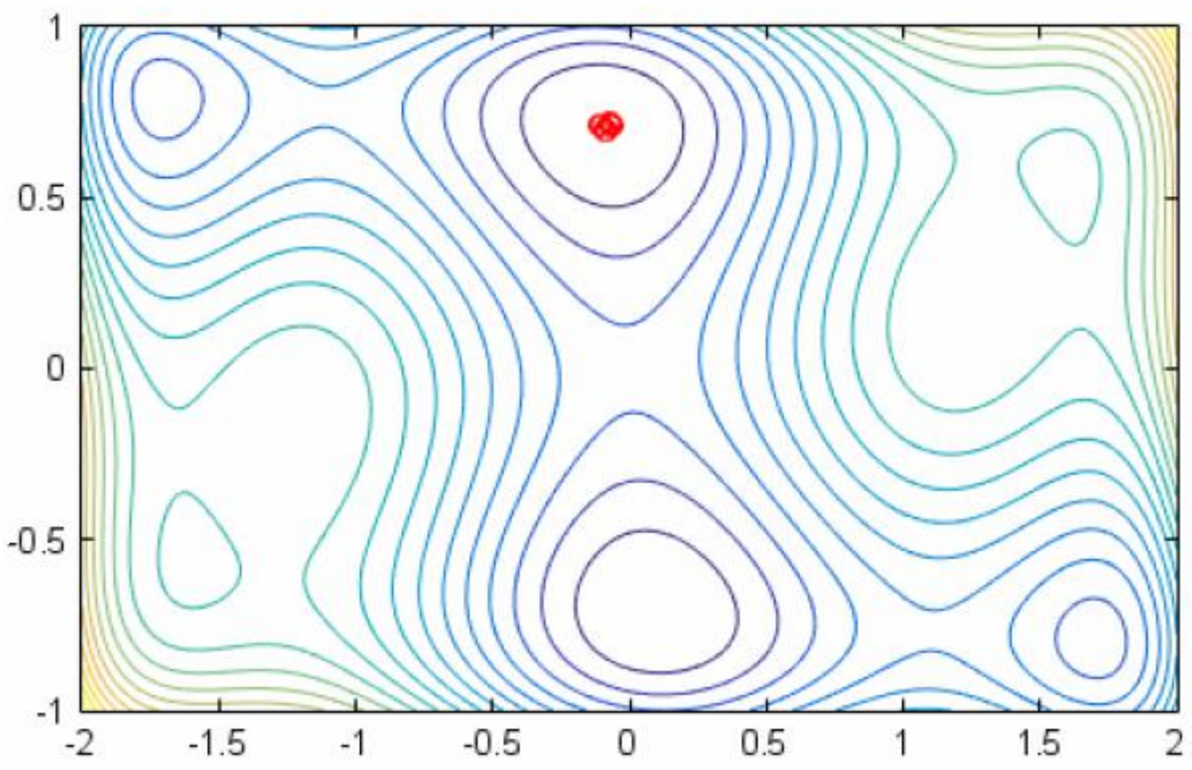
Figures and slide concept: Simon Hansul

- // L0 (initial length) added as a parameter
- // Initial visual fit to the data resulted in the following parameter set:
  rB = 0.2
  L_∞ = 2.5
  L_0 = 1
- // A pragmatic solution for dealing with replicates is using the mean of the data (over time)

Applying a calibration algorithm

- // Find parameters with minimum loss (optimization) algorisms such as
  - // Nelder-Mead
  - // Maximum likelihood estimation

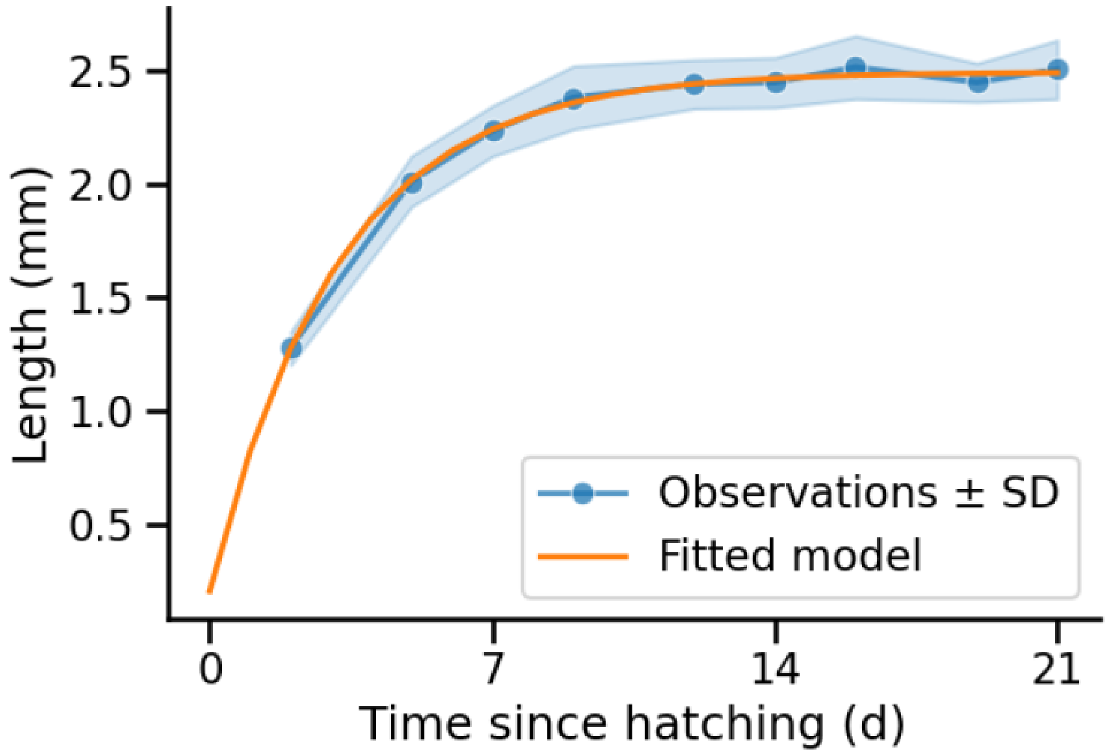
- // Or construct probability distributions via Bayesian inference, e.g.
  - // Markov-Chain Monte-Carlo
  - // Approximate Bayesian Computation (ABC)



Figures and slide concept: Simon Hansul

Model fitting example

Model fit using Nelder-Mead



Figures and slide concept: Simon Hansul

- Resulting paramter estimates are:
  - rB = 0.31
  - L_∞ = 2.1
  - L_0 = 0.21
- Important diagnostic steps:
  - Visual predictive check
  - Quantitative predictive check
  - Interpretation of parameter values
  - Plausability checks (e.g. comparison with existing studies and simple extrapolations)

Back to exercises

Exercise 3

Use the cadmium bioaccumulation data for parameter estimation

// Estimate the parameters k_u and k_E by fitting the one-compartment bioaccumulation model to the data using the *rbioacc* package in R

// First, curate the data in accordance with the package requirements. For an example see Table 1 in Ratier et al. 2022

// Read data in R and execute the fitting routine of the package (you may find the code example on the next slide hepful)

// Plot fitting results and get parameter estimates (see code example)

// Determine the bioconcentration factor from kinetics (BCF $=k_u/k_e$)

Code example for running the `rbioacc` package

```
library("rbioacc")

data <- read.csv("bioacc_data.csv", header = T, sep = ",", dec = ".") # data

# specify the time at which the accumulation phase ends (e.g.80 hours):
modeldata <- modelData(data, time_accumulation = 80)

# fit the TK model
m <- fitTK(modeldata)

# plot the fitting results
plot(m)
q<-quantile_table(m)
q[1:2,]
```

Exercise 4

// Use your implementation of the the one-compartment bioaccumulation model from Exercise 1.

// Simulate bioaccumulation for a variable exposure scenario and for constant exposure of the area under the curve concentration over (exposure_scenario.csv) time using the parameters estimated in Exercise 3.

// Show model results for both scenarios in a graph.

// What is the difference between variable exposure and constant exposure in terms of bioaccumulation?

Exercise 5

- // Use your implementation of the the one-compartment bioaccumulation model
- // Replace/add the bioaccumulation model by/to the one-parameter function for the scaled internal concentration C_i:

$$\frac{dC_i}{dt} = k_d(C_w - C_i)$$

- // How is this function related to the one compartment model presented on Slide 18? Which assumption do you need to make to derive a value for k_d to simulate Cadmium bioaccumulation in *D. magna*?
- // Compare modeling results for the scaled internal concentration with the ones obtained from Exercise 4